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A Simulation-Based FMU Digital-Twin Workflow for Fault-Informed Thermal Supervisory Control of an Induction Motor

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Abstract

This study presents a simulation-based digital-twin workflow for an 18.5 kW squirrel-cage induction motor composed around the Modelica Standard Library benchmark and exported as a Functional Mock-up Unit (FMU) for Functional Mock-up Interface (FMI) 2.0 Co-Simulation. A Python loop integrates thermal-state estimation, fault diagnosis, 30 s and 60 s temperature forecasting, three supervisory controllers, and an Asset Administration Shell representation. Nominal behavior was checked against the Modelica reference example; the study is not a physical-motor validation. A deterministic campaign covered 11 scenarios and three controllers (33 runs; 2700 s per run; 0.5 s sampling). No run crossed the configured critical policy threshold of 120 degrees C. Scenario-mean maximum winding temperatures were 96.171, 95.343, and 94.825 degrees C for baseline, adaptive, and constrained control, respectively; corresponding mean useful shaft work was 12.377, 13.104, and 13.278 kWh, while electrical energy was 13.639, 14.414, and 14.602 kWh. These deterministic simulation outcomes show policy trade-offs, not statistical generalization or validated protection performance. Thermal parameters and fault magnitudes remain assumptions, and the model has not been validated on physical hardware.

Keywords: induction motor; digital twin; thermal estimation; fault diagnosis; Modelica; FMU; supervisory control; predictive maintenance

1. Introduction

This paper documents a simulation-based digital-twin framework for a squirrel-cage induction motor built from the Modelica Standard Library (MSL) machine model and evaluated under a deterministic multi-scenario campaign. The active plant is an 18.5 kW, 400 V, 50 Hz, four-pole induction machine composed around the unmodified MSL squirrel-cage machine component. The project integrates the plant with a lumped winding/frame thermal network, a Functional Mock-up Unit (FMU) exported for Functional Mock-up Interface (FMI) 2.0 Co-Simulation, a Python runtime, a thermal-state estimator, fault diagnosis and severity estimation, 30 s and 60 s forecasting, and supervisory control. The system is not presented as a physical motor-test validation; rather, it is a simulation benchmark for digital-twin experimentation under controlled fault scenarios.

Digital-twin terminology spans models linked to physical assets, model-based digital shadows, and simulation-oriented workflows; these categories are not interchangeable

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and depend on the direction and timing of data exchange [1]. Induction-motor condition monitoring has a substantial literature on current-signature analysis and comparative fault-diagnosis methods [2,3]. More recent motor-focused digital-twin studies address particular diagnostic settings: Adamou and Alaoui describe efficiency- and loss-based diagnosis for broken-rotor-bar cases, while Chen et al. report a digital-twin method for stator inter-turn short circuits in traction motors [4,5]. The available publisher metadata and abstracts support these descriptions; full-text methods/results could not be independently reviewed here, so no numerical performance comparison is made. Neither study is treated as a direct controller benchmark for the present work.

The thermal network used here is a lumped engineering approximation; established electric-machine thermal analysis emphasizes that heat paths and boundary conditions matter and that model fidelity depends on the intended application [6]. The present work is therefore an engineering integration and deterministic simulation-evaluation study, not a field-validated industrial deployment or an algorithmic novelty claim. Its bounded contributions are (i) an MSL-based induction-machine FMU linked to an explicit Python estimator/diagnosis/forecast/control loop, (ii) an auditable 11-scenario comparison with true plant values kept separate from Twin estimates, and (iii) a structured six-submodel Asset Administration Shell (AAS) synchronization demonstration. The research questions are: (1) how closely does the composed plant reproduce the MSL nominal operating point under the documented test, and (2) what deterministic trade-offs in thermal outcomes, useful shaft work, energy, and prediction errors are observed across the three implemented controllers? These questions are answerable only for this model and protocol.

The remainder of the paper is organized as follows. Section 2 describes the plant model, parameter provenance, thermal network, FMU pipeline, and evaluation protocol. Section 3 reports the nominal operating-point validation and the 33-run deterministic campaign. Section 4 interprets the results, discusses limitations, and positions the work against prior literature. Section 5 summarizes the contributions and identifies future validation needs.

2. Materials and Methods

2.1. System architecture and scope

The simulated plant composes the unmodified MSL squirrel-cage induction-machine component with a lumped winding/frame thermal network, estimator-facing measurement channels, fault-injection inputs, and supervisory-control interfaces. The model is exported to FMI 2.0 Co-Simulation and executed by a Python runtime that advances the FMU in communication substeps no larger than 0.1 s. True plant variables, estimates, predictions, and controller commands are recorded separately at sample intervals of 0.5 s.

The architecture follows a single closed-loop digital-twin workflow: plant state → measurement adaptation → thermal estimation → fault detection/severity estimation → forecast generation → controller actuation → plant update (Figure 1). The controller logic is intentionally heterogeneous: baseline control relies on direct thermal condition and threshold logic, whereas adaptive and constrained control also use observer, anomaly, and forecast information. This difference in information access is preserved in the campaign and should be considered when comparing controller outcomes.

The implementation exposes six AAS submodels—OperationalState, ElectricalState, MechanicalState, ThermalState, FaultState, and ControlInterface—following the structured submodel approach in the AAS specification [9]. At each synchronization, a shared snapshot identifier and simulation timestamp are written and read back across the six submodels. This local BaSyx test establishes the tested write/read/alignment behavior only; it is not evidence of an industrial deployment.

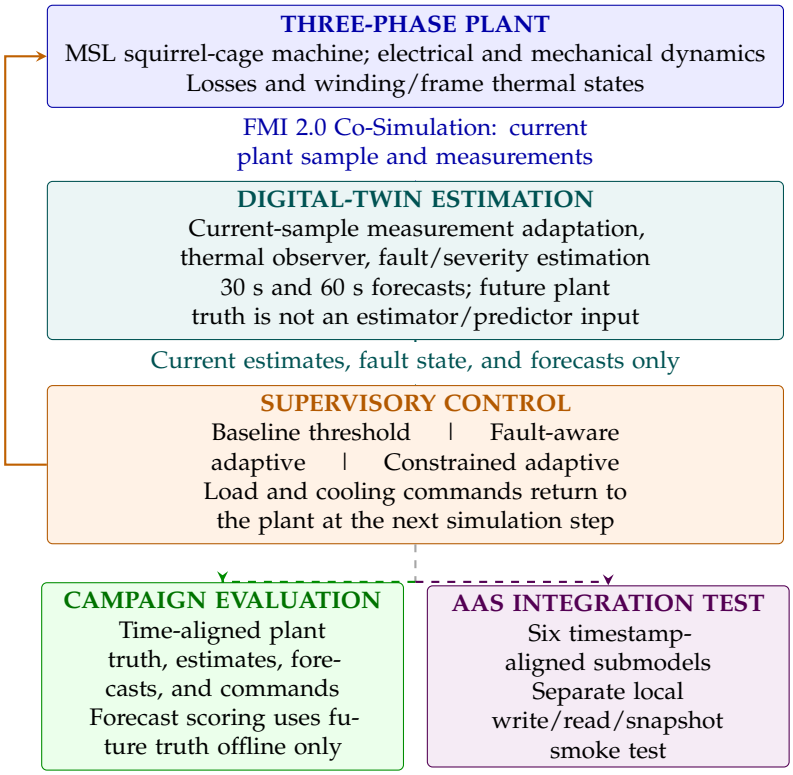


Figure 1. Simulation workflow and evidence paths. Plant truth is recorded separately from Twin estimates; future plant values are used only for post-run forecast scoring and are not estimator or predictor inputs. The Asset Administration Shell (AAS) branch represents a separate local integration test, not a physical deployment.

2.2. Plant model and parameter provenance

The selected benchmark is the MSL `IMC_withLosses` example in MSL 4.1.0 [7], which provides a published 18.5 kW, 400 V, 50 Hz, four-pole squirrel-cage induction-machine parameter set and associated nominal operating point. The project does not claim that the benchmark is a manufacturer-identified motor or a field-calibrated machine. Instead, the model is an MSL-based benchmark whose nominal values include rated shaft output, line-to-line RMS voltage, nominal current, efficiency, speed, and loss data.

Table 1 summarizes the model attributes and parameter provenance. Values were obtained from (i) MSL reference data, (ii) literature/reference values, (iii) calculations from documented values and formulas, and (iv) explicit assumptions for thermal and controller configuration. The stator and rotor resistances, leakage inductances, magnetizing inductance, inertia, and loss quantities are inherited from the MSL benchmark or derived from its documented nominal conditions. The winding/frame thermal capacitances and conductances, fault magnitudes, safety thresholds, and controller envelopes are assumptions rather than manufacturer or experimental measurements.

Table 1. Main induction-motor model attributes and provenance. MSL-derived electrical and mechanical values are distinguished from assumed or calculated thermal-network parameters.

Attribute	Value	Provenance
Rated shaft output	18.5 kW	MSL reference data
Supply voltage	400 V line-to-line RMS	MSL reference data
Frequency	50 Hz	MSL reference data
Pole count	4 poles	MSL parameter-record definition
Nominal speed	1462.5 rpm	MSL reference data
Nominal efficiency	90.49%	MSL reference data
Stator and rotor resistances	0.56 Ω/phase, 0.42 Ω/phase	MSL reference data
Magnetizing/leakage inductances	Derived from MSL reactances	Calculated values
Rotor inertia	0.12 kg m^2	MSL reference data
Ambient temperature	20 °C	Explicit assumption
Winding heat capacity	15,000 J K^{-1}	Assumed thermal parameter
Frame heat capacity	35,000 J K^{-1}	Assumed thermal parameter
Winding-to-frame resistance	0.010 K W^{-1}	Assumed thermal parameter
Frame-to-ambient resistance	0.0296652 K W^{-1}	Calculated from MSL loss/reference temperatures and assumed network
Winding/fixed loss split	0.644/0.356	Calculated from MSL nominal copper and total losses
Thermal safety thresholds	90/100/115/120 °C policy	Digital-twin configuration

The nominal validation checks the MSL reference operating point rather than a physically instrumented motor. Under balanced nominal three-phase supply, shaft load is ramped to nominal torque and the settled model outputs are compared against the MSL reference values. Voltage, current, frequency, speed, slip, electromagnetic torque, shaft torque, electrical input power, shaft power, efficiency, power factor, and loss components are checked with explicit tolerances. This is a benchmark consistency check, not hardware validation.

2.3. Thermal model and fault layer

The plant's winding and frame nodes receive losses from the MSL machine thermal ports. With winding copper loss P_w, remaining internal machine loss P_f, and effective frame-to-ambient resistance $R_{fa,eff}$, the implemented two-node balance is

$$C_w \frac{dT_w}{dt} = P_w - \frac{T_w - T_f}{R_{wf}}, \qquad C_f \frac{dT_f}{dt} = P_f + \frac{T_w - T_f}{R_{wf}} - \frac{T_f - T_a}{R_{fa,eff}}. \tag{1}$$

Here, $C_w = 15{,}000$ J K^{-1}, $C_f = 35{,}000$ J K^{-1}, and $R_{wf} = 0.010$ K W^{-1} are assumptions. The nominal frame-to-ambient resistance is 0.0296652 K W^{-1}, derived from the selected nominal loss and temperature values under the assumed network. Cooling and thermal-resistance degradation modify effective heat rejection within configured bounds. The model routes winding losses to the winding node and the other MSL thermal-port losses to the frame node; the resulting network does not resolve internal spatial gradients or all physical heat paths. Additional externally imposed friction work is included in the total loss output but is not a separately identified measured thermal input. These dynamics are a model approximation, not a calibrated motor thermal model.

The fault layer applies explicit scenario disturbances to the supply amplitude or frequency, one phase voltage, cooling/thermal resistance, sensor channel, shaft load, or opposing mechanical torque. Table 2 summarizes the 11 tested scenarios and their modeled mechanisms. Mechanical unbalance and radial vibration are not represented because the plant has no radial mechanical degrees of freedom. Fault amplitudes are stress-test inputs, not calibrated field severities.

Table 2. Deterministic campaign scenarios and the modeled intervention. The mechanisms describe simulation inputs, not validated field fault signatures.

Scenario	Modeled intervention
healthy	No injected fault; nominal reference condition
cooling	Reduced frame-to-ambient cooling effectiveness
frequency deviation	Supply frequency shifted from nominal
mechanical friction	Additional speed-opposing mechanical torque
overload	Increased shaft-load torque
thermal-resistance degradation	Increased winding-to-frame thermal resistance
sensor bias	Additive bias to the winding-temperature sensor channel
sensor freeze	Hold the sensor output at its value at fault activation
supply degradation	Scale down balanced phase-voltage amplitudes
voltage imbalance	Modify one phase voltage to create supply imbalance
combined supported	Combined sensor bias, cooling degradation, overload, and supply-voltage reduction

2.4. FMU runtime, estimation, forecasting, and control

The FMU runtime advances the exported model using FMI 2.0 Co-Simulation [8], converts outputs to the controller interface, and stores true plant channels separately from estimates, forecasts, and commands. The estimator uses current loss/temperature measurements and an internal two-node model; it initializes from the current temperature measurements, propagates winding temperature from the loss balance, and corrects the frame state using its measured innovation. It does not correct the winding estimate directly from the potentially biased/frozen winding sensor. The predictor rolls the same reduced-order model forward in 0.5 s steps to 30 s and 60 s, holding current measured loss inputs and the current cooling command/effectiveness constant over each forecast horizon. Future FMU truth is used only after the run to score the forecasts, not as an estimator/controller input.

The prediction block estimates 30 s and 60 s winding-temperature trajectories from current plant and observer data. The controller block implements a baseline threshold policy, a fault-aware adaptive policy, and a constrained adaptive policy. Table 3 records the respective inputs. The baseline policy uses raw winding-sensor temperature and detector-derived sensor reliability; the adaptive policy uses the state estimate, detector, severity scores, and forecasts; the constrained policy uses estimated and predicted temperatures, overall fault severity, inferred cooling efficiency, and a sensor-reliability factor. Controller signal access is therefore unequal, and the comparison is an engineering trade-off study rather than a matched algorithm benchmark.

The supervisory `speed_pu` input scales the supply frequency command within the model; it is not a closed-loop shaft-speed setpoint. The reported shaft speed is an emergent plant output under the electrical supply and load torque. Speed tracking errors are therefore interpreted relative to the nominal V/f reference used by the metric, not as tracking performance of a speed controller.

Table 3. Controller information used by the implemented supervisory policies.

Controller	Inputs used for its command	Implemented policy
Baseline	Raw winding sensor; reliability flag from sensor-bias/freeze detection	Threshold derating at warning/critical temperatures; minimum-load command if sensor is unreliable
Adaptive	Estimated winding state; fault detection; severity channels; 30 s and 60 s forecasts	Forecast-/risk-weighted load derating with fault penalties and per-step load-rate limit
Constrained	Estimated and forecast winding temperatures; overall fault severity; inferred cooling efficiency; sensor reliability	Prioritizes cooling, then constrained progressive load derating near risk limits

2.5. Evaluation protocol and reproducibility

The final deterministic campaign comprises 11 scenarios and three controllers, yielding 33 runs. Each run lasts 2700 s with fault inputs applied from 350 s and 0.5 s control samples. The sample at 350 s is pre-injection; the first faulted sample is at 350.5 s. No stochastic disturbance or random seed is used. Results are computed from raw per-run CSV files and the manifest. Electrical energy is left-endpoint integrated, useful work integrates only positive shaft power, and mean scenario outcomes weight the 11 scenarios equally. Estimation errors compare same-time estimates with FMU truth; forecast errors compare predictions with future truth only during post-run scoring, omitting horizons without a matching future sample. The equations are

$$\begin{aligned} E_{\mathrm{e}} &= \frac{\sum_i P_{\mathrm{e}}(t_i)\Delta t_i}{3.6\times 10^6}, \qquad W_{\mathrm{shaft}} = \frac{\sum_i \max(P_{\mathrm{shaft}}(t_i),0)\Delta t_i}{3.6\times 10^6}, \\ \mathrm{MAE}_h &= \frac{1}{N_h}\sum_{i\in I_h}\left|\widehat{T}_{\mathrm{w},h}(t_i) - T_{\mathrm{w}}(t_i+h)\right|, \end{aligned} \tag{2}$$

where powers are in watts, Δt_i is in seconds, energy and work are in kWh, $h \in \{30, 60\}$ s, and I_h contains only samples with an aligned target at horizon h. The AAS write/read/snapshot smoke test is a separate run. The campaign manifest records Python 3.12.4, Windows 11 (build 22621), NumPy 2.5.3, pandas 3.0.6, FMPy 0.3.32, and Matplotlib 3.11.2. OpenModelica 1.27.1 and MSL 4.1.0 were used for FMU export and nominal validation; detailed hardware model/RAM information was not recorded. The campaign manifest records source commit `d14c35ff2e73ebf743c970f7585b3c3030140b97` and a non-clean working tree; the exact source delta therefore still needs a versioned archival snapshot before independent reproduction. From the repository root, the documented campaign and artifact-regeneration commands are:

```
.\.venv\Scripts\python.exe -m experiments.run_research_evaluation
.\.venv\Scripts\python.exe -m experiments.run_research_evaluation `
  --with-basyx
.\.venv\Scripts\python.exe -m analysis.generate_research_artifacts `
  --campaign results\research_evaluation\<UTC timestamp>
```

For the separate local AAS smoke test, the documented service launch and seed commands are:

```
docker compose -f docker\docker-compose.yml up -d
.\.venv\Scripts\python.exe scripts\seed_basyx.py
```

The optional `-with-basyx` campaign mode is separate from the completed AAS smoke-test evidence. No random seed or stochastic input is used.

3. Results

3.1. Nominal validation

The nominal validation step compares the model against the benchmark MSL reference operating point for a balanced 400 V, 50 Hz, nominal-load condition. All explicit tolerance checks passed under OpenModelica 1.27.1 and MSL 4.1.0. Table 4 reports the observed settled values and signed deviation (simulation minus reference) for the primary operating-point variables. The run demonstrates nominal consistency with the MSL benchmark point, not independent physical validation of an identified motor.

Table 4. Nominal operating-point check against the MSL reference. Deviations are simulation output minus reference; the validation JSON reports that each measure passed its configured tolerance.

Quantity	Reference	Simulation	Deviation
Line voltage (V)	400.000	400.000	< 0.001
Line current (A)	32.850	32.857	+0.0065
Supply frequency (Hz)	50.000	50.000	0.000
Shaft speed (rpm)	1462.500	1462.894	+0.394
Slip (%)	2.500	2.474	−0.026 percentage points
Electromagnetic torque (N m)	122.637	122.638	+0.0009
Shaft torque (N m)	120.795	120.795	< 0.0001
Electrical input power (kW)	20.444	20.444	+0.00036
Shaft output power (kW)	18.500	18.505	+0.00498
Efficiency (%)	90.490	90.514	+0.024 percentage points
Power factor	0.8980	0.8981	+0.00011
Total motor losses (W)	1943.950	1939.330	−4.620

3.2. Deterministic campaign overview

The final campaign executed 33 completed runs, with no failed runs. Raw traces, a campaign manifest, and summary metrics provide the underlying evidence. Table 5 reports the mean values across the equally weighted scenario set for the three controllers. These are deterministic scenario averages, not statistical estimates of population-level motor performance or field behavior.

Table 5. Mean deterministic campaign values over the 11 equally weighted scenarios for each controller. Values should be interpreted as scenario-level averages rather than statistical estimates.

Metric	Baseline	Adaptive	Constrained
Mean maximum winding temperature	96.171 °C	95.343 °C	94.825 °C
Mean useful shaft work	12.377 kWh	13.104 kWh	13.278 kWh
Mean electrical energy	13.639 kWh	14.414 kWh	14.602 kWh
Mean winding-estimation MAE	1.131 °C	0.943 °C	1.036 °C
30 s forecast MAE	0.934 °C	0.721 °C	0.819 °C
60 s forecast MAE	1.022 °C	0.764 °C	0.869 °C

Figure 2 shows representative winding-temperature trajectories and illustrates that no configured run exceeded the 120 °C critical threshold. The smallest minimum thermal safety margin was 7.259 °C in the baseline cooling-degradation case, in which the maximum winding temperature reached 112.741 °C. This result confirms only that the simulated traces remained below the configured threshold; it does not validate a manufacturer-specific thermal limit or motor protection boundary.

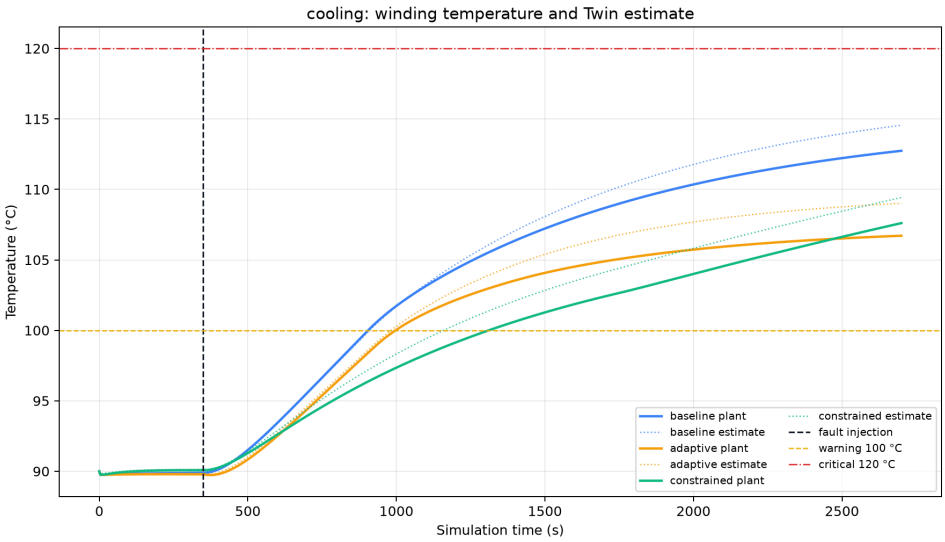


Figure 2. Winding-temperature truth and Twin estimates for the cooling-degradation scenario for all three controllers. Fault input is applied at 350 s; the sample at 350 s is pre-fault and the first faulted sample is at 350.5 s. The 100 °C warning and 120 °C critical lines are configured policy thresholds, not validated motor safety limits.

Figure 3 shows the winding-temperature response and Twin estimates for voltage imbalance. It provides a second fault case in which controller behavior and estimator-to-plant differences can be inspected directly; it does not establish physical diagnostic accuracy.

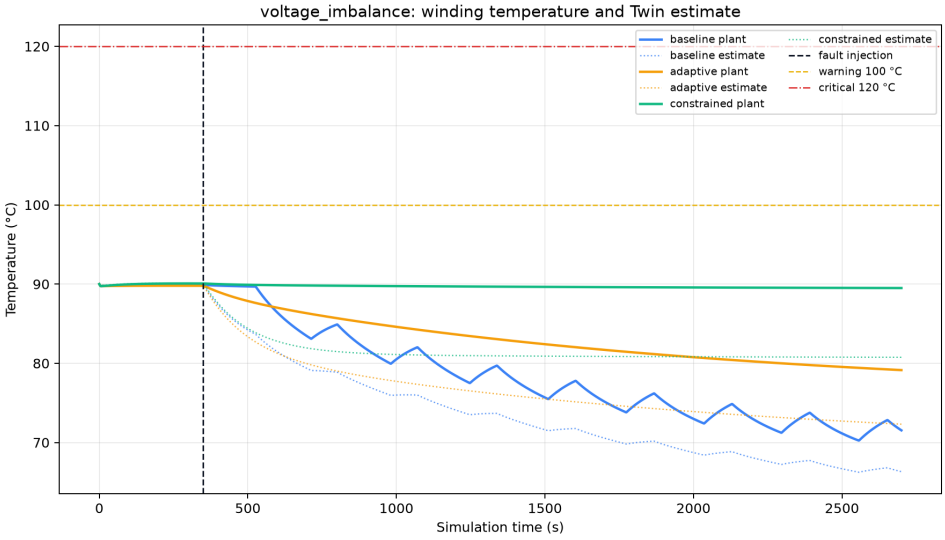


Figure 3. Winding-temperature plant truth and Twin estimates for the voltage-imbalance scenario. The fault input is applied at 350 s; the 100 °C and 120 °C lines denote configured warning and critical policy thresholds, not validated motor safety limits. The discrepancy among plant and estimate trajectories illustrates this simulated case only.

Figure 4 compares the three supervisory load commands in the cooling-degradation scenario. The commands diverge after fault activation because the policies use different measurements and estimated/predicted information, as summarized in Table 3.

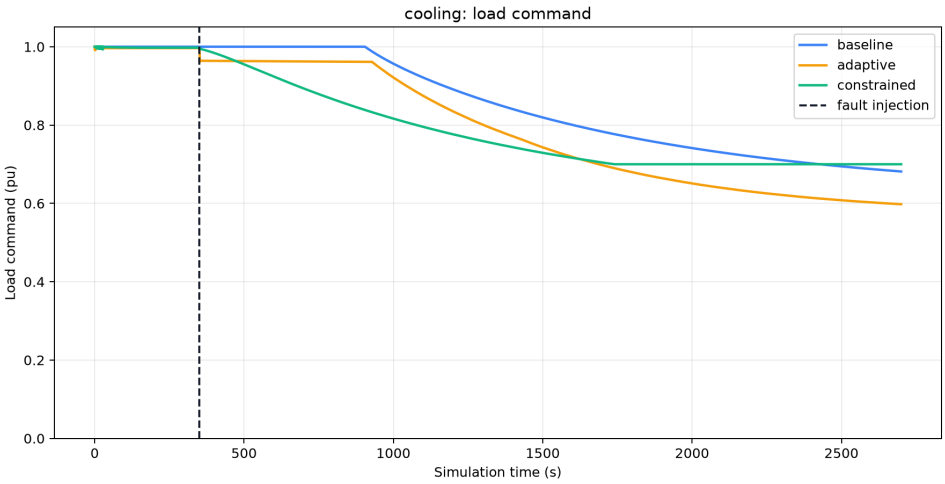


Figure 4. Normalized load command during the cooling-degradation scenario for the baseline, adaptive, and constrained policies. The vertical dashed line marks fault injection at 350 s. This trace documents the implemented commands in one deterministic run and is not evidence of a generally optimal control response.

3.3. Estimation and forecasting performance

Winding-estimation mean absolute error (MAE) and 30 s/60 s forecast MAE were evaluated for each controller. On the deterministic scenario-average basis, adaptive control has the lowest average forecast MAE; constrained control has the lowest mean maximum winding temperature and the highest mean useful shaft work, together with higher electrical energy than baseline. These outcomes are not a universal ranking: the controllers use different information and policies, and each scenario/controller pair is one deterministic run. Figure 5 presents scenario-level forecast error and illustrates substantial variation among the tested scenarios.

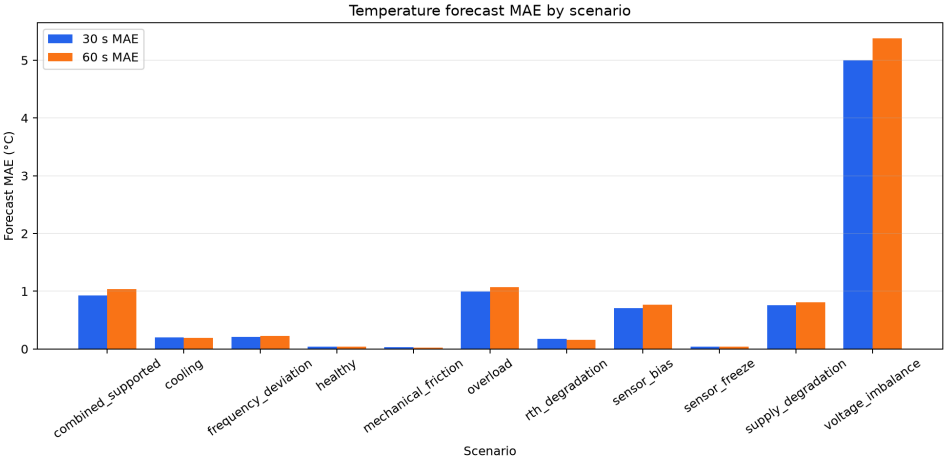


Figure 5. Temperature forecast mean absolute error by scenario at 30 s and 60 s. Values are deterministic scenario-level errors computed from aligned future truth samples; they are not repeated-run uncertainty estimates. The voltage-imbalance case has substantially larger errors than the other tested cases.

3.4. BaSyx smoke test

The live BaSyx AAS workflow was evaluated separately from the main deterministic campaign. The smoke test recorded one successful write, 12 successful reads, and one aligned snapshot across the six named submodels. The measured end-to-end synchronization loop took 1055.599 ms and write latency was 443.361 ms; mean read latency was 50.623

ms with a reported p95 of 133.255 ms across the observed reads. The single write and single snapshot are not a statistically meaningful latency distribution. These values demonstrate local write/read/snapshot behavior only, not a real-time guarantee or industrial deployment.

4. Discussion

This study provides an engineering integration contribution, not a claim of physical motor certification. The FMU-based workflow demonstrates an end-to-end digital-twin pipeline with an MSL induction-machine benchmark. The implemented workflow connects Modelica plant modeling to co-simulation, observer/forecast logic, scenario-based fault injection, and deterministic controller comparison. It also demonstrates how AAS submodel semantics can be integrated into a digital-twin loop. These contributions should not be overstated as proven field performance or industrial deployment.

Prior work establishes that induction-motor diagnosis spans current-signature methods, fault-specific digital-twin models, and thermal-analysis approaches [2–6]. This project does not claim a new diagnosis or control algorithm and does not compare against those studies' diagnostic benchmarks. Its contribution is an implementation/evaluation artifact: a composed MSL machine with explicit thermal and fault interfaces, a Python loop that separates current Twin estimates from simulation truth, and a deterministic report of thermal, energy, useful-work, and prediction outcomes. The MSL benchmark is not an identified commercial motor, the thermal network is assumption-based, and fault severities are synthetic scenario inputs rather than measured industrial degradation.

The results also reveal several fundamental limitations. The plant is an MSL example-based model, not a manufacturer-identified or physically calibrated motor. Thermal capacitances and resistances are assumed or calculated from assumed network structure; the two-node network aggregates core and other machine losses and omits detailed thermal paths and spatial gradients. External degradation-friction work is not separately coupled as heat to the network. Fault magnitudes, residual thresholds, and 100/120 °C warning/critical values are not validated against equipment. Residual thresholds were tuned using simulated channels, and the model does not reproduce measured sensor noise/dynamics. Mechanical unbalance and radial vibration are not modeled. Finally, the 33 runs are deterministic single runs with no stochastic replication, so the results do not estimate uncertainty or real-motor generalization.

The controller comparison is also only partially comparable because the system does not give all controllers identical signal access. The baseline controller uses a more limited set of thermal and operating indicators, while the adaptive and constrained controllers additionally exploit observer and forecast outputs. Therefore, the results should be read as an engineering trade-off study rather than as a universal statement that one controller is superior. The adaptive controller shows the best average forecast accuracy and strong thermal safety, while the constrained controller is competitive on useful work and thermal safety, but no universal ranking should be inferred beyond the tested deterministic scenarios.

5. Conclusions

This manuscript documents a simulation-based digital-twin workflow for an MSL-based induction motor with thermal-state estimation, simulated fault diagnosis, forecasting, and supervisory control. The evidence supports three bounded conclusions: the composed 18.5 kW model is nominally consistent with the selected MSL reference point under the stated test; the final deterministic campaign has no simulated trace at or above the configured 120 °C threshold; and the three policies produce distinct scenario-average outcomes

in thermal temperature, useful shaft work, electrical energy, and forecast error. None of these results establishes physical fault-detection performance, manufacturer-specific safety, real-time capability, or industrial deployment.

The project does not claim field validation, manufacturer calibration, real-time certification, or general industrial deployment. The work is better described as a transparent and reproducible simulation-based digital-twin engineering framework with explicit limitations and clearly separated evaluation evidence. Future work should prioritize plant-level experimental calibration, measured thermal parameter identification, updated diagnostic thresholds from physical data, and broader validation studies that compare digital-twin estimates against instrumented motor tests.

Author Contributions: Software, investigation, and writing—original draft preparation, D.A.; supervision and writing—review and editing, S.R. The corresponding author reports that both authors reviewed the prior draft and agreed to submission after the requested revisions; the authors should confirm approval of this revised version before submission.

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Institutional Review Board Statement: Not applicable. This study used numerical simulation and did not involve human participants or animals.

Informed Consent Statement: Not applicable.

Data Availability Statement: The project repository is available at https://github.com/Dilshad-Ahmed1/Industrial-Motor-DT. It provides project source and documentation, but the exact source state used for the reported campaign and its raw traces have not been deposited as a versioned public archive. The campaign manifest identifies a source commit but also records a non-clean working tree; the commit alone does not reproduce the reported study. Before submission, the authors should archive the exact source delta, configuration, raw traces, environment record, and analysis materials or provide an author-approved access arrangement.

Supplementary Materials: No supplementary files are currently deposited. A reproducibility archive containing the exact campaign source state and raw traces remains to be prepared before submission.

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Conflicts of Interest: The authors declare no conflicts of interest.

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